^{24}Mg(^{14}N,3pγ) 2014Ay01

2014Ay01: ^{24}Mg(^{14}N,3p)^{35}S fusion-evaporation reaction. A 40-MeV, 5-pnA ^{14}N beam was delivered from the LNL XTU-Tandem accelerator and impinged on 1-mg/cm^2 thick, 99.7% isotopically enriched ^{24}Mg evaporated on an 8-mg/cm^2 gold layer. γ rays were detected using 4π-GASP array composed of 40 Compton-suppressed large volume HPGe detectors arranged in seven rings at different angles with respect to beam axis. Events were collected when at least two germanium detectors fired in coincidence. Measured Eγ, Iγ, $\gamma\gamma$-coin, $\gamma(\theta)$, $\gamma\gamma(\theta)$(ADO). Deduced levels, J, π, level lifetimes via the Doppler shift attenuation method (DSAM). Compared with large-scale shell model calculations.

^{35}S Levels

E(level)[†]	J$^\pi$[‡]	E(level)[†]	J$^\pi$[‡]	T$_{1/2}$[#]	E(level)[†]	J$^\pi$[‡]	T$_{1/2}$[#]
0.0[&]	3/2$^+$	3816.0[@] 3	9/2$^-$	0.28 ps 3	5412.2 3	9/2$^+$	
1572.5 3	1/2$^+$	3886.4 4	5/2$^+$		5877.6[&] 3	11/2$^+$	
1991.4[@] 2	7/2$^-$	4023.2[@] 3	11/2$^-$	0.32 ps 3	6299.4 4	11/2$^+$	
2347.9 3	3/2$^-$	4822.9 3	9/2$^+$		6352.2[@] 5	13/2$^-$	50 fs 10
2717.1 3	5/2$^+$	4899.7 4	9/2$^+$		7179.8[&] 3	15/2$^+$	3.1 ps 12
3594.7[&] 3	7/2$^+$	5010.2 3	11/2$^-$	0.45 ps 8	8023.7[&] 9	17/2$^+$	0.15 ps 4

† From a least-squares fit to γ-ray energies.
‡ As given in 2014Ay01, which quotes the known J^π assignments for four low-lying levels at 1572, 1991, 2347, and 2717 from 2011Ch48. 2014Ay01 assigns J^π for levels above 3 MeV based on the measured $\gamma\gamma(\theta)$(ADO) ratios. When considered in the Adopted Levels, the firm assignments here are placed within parentheses if there are no other strong arguments to support these firm assignments.
# From DSAM line-shape analyses (2014Ay01).
@ Seq.(A): ΔJ=1 sequence based on 7/2$^-$.
& Seq.(B): Sequence based on 3/2$^+$.

γ(^{35}S)

R$_{ADO}$=[Iγ(34°)+Iγ(146°)]/2Iγ(90°). Expected values are 0.8 for stretched dipole (ΔJ=1) and 1.4 for stretched quadrupole (ΔJ=2) or unstretched dipole (ΔJ=0) transitions.

E$_\gamma$	I$_\gamma$	E$_i$(level)	J$^\pi_i$	E$_f$	J$^\pi_f$	Mult.[†]	Comments
207.0 5	0.4 2	4023.2	11/2$^-$	3816.0	9/2$^-$		
465.3 3	1.2 1	5877.6	11/2$^+$	5412.2	9/2$^+$	D	R$_{ADO}$=0.71 9.
775.4 4	1.4 2	2347.9	3/2$^-$	1572.5	1/2$^+$	D	R$_{ADO}$=0.68 14.
827.8 6	0.3 1	7179.8	15/2$^+$	6352.2	13/2$^-$		
843.9 8	1.6 2	8023.7	17/2$^+$	7179.8	15/2$^+$	D	R$_{ADO}$=0.8 3.
867.3 3	1.9 1	5877.6	11/2$^+$	5010.2	11/2$^-$	D	R$_{ADO}$=1.36 20, consistent with ΔJ=0.
877.6 5	0.4 1	3594.7	7/2$^+$	2717.1	5/2$^+$		
880.2 4	0.8 1	7179.8	15/2$^+$	6299.4	11/2$^+$		
887.0 7	0.4 1	6299.4	11/2$^+$	5412.2	9/2$^+$	D	R$_{ADO}$=0.6 3.
977.8 3	1.6 1	5877.6	11/2$^+$	4899.7	9/2$^+$	D+Q	R$_{ADO}$=1.43 16.
986.9 3	4.4 2	5010.2	11/2$^-$	4023.2	11/2$^-$	D	B(E1)(W.u.)=0.00102 +22−16; B(M1)(W.u.)=0.036 +8−6 R$_{ADO}$=1.30 13, consistent with ΔJ=0.
1006.8 2	0.8 1	4822.9	9/2$^+$	3816.0	9/2$^-$	D	R$_{ADO}$=1.37 11, consistent with ΔJ=0.
1054.5 3	3.0 2	5877.6	11/2$^+$	4822.9	9/2$^+$	D	R$_{ADO}$=0.72 11.
1228.1 3	1.1 1	4822.9	9/2$^+$	3594.7	7/2$^+$	D	R$_{ADO}$=0.78 9.
1302.2 2	13.5 8	7179.8	15/2$^+$	5877.6	11/2$^+$	E2	B(E2)(W.u.)=6.6 +42−19 R$_{ADO}$=1.40 5, A$_2$=+0.27 5, A$_4$=−0.07 5.
1304.8 12	0.6 2	4899.7	9/2$^+$	3594.7	7/2$^+$		
1389.0 4	3.9 4	5412.2	9/2$^+$	4023.2	11/2$^-$	D	R$_{ADO}$=0.79 11.

Continued on next page (footnotes at end of table)

^{24}Mg(^{14}N,3pγ) 2014Ay01 (continued)

γ(^{35}S) (continued)

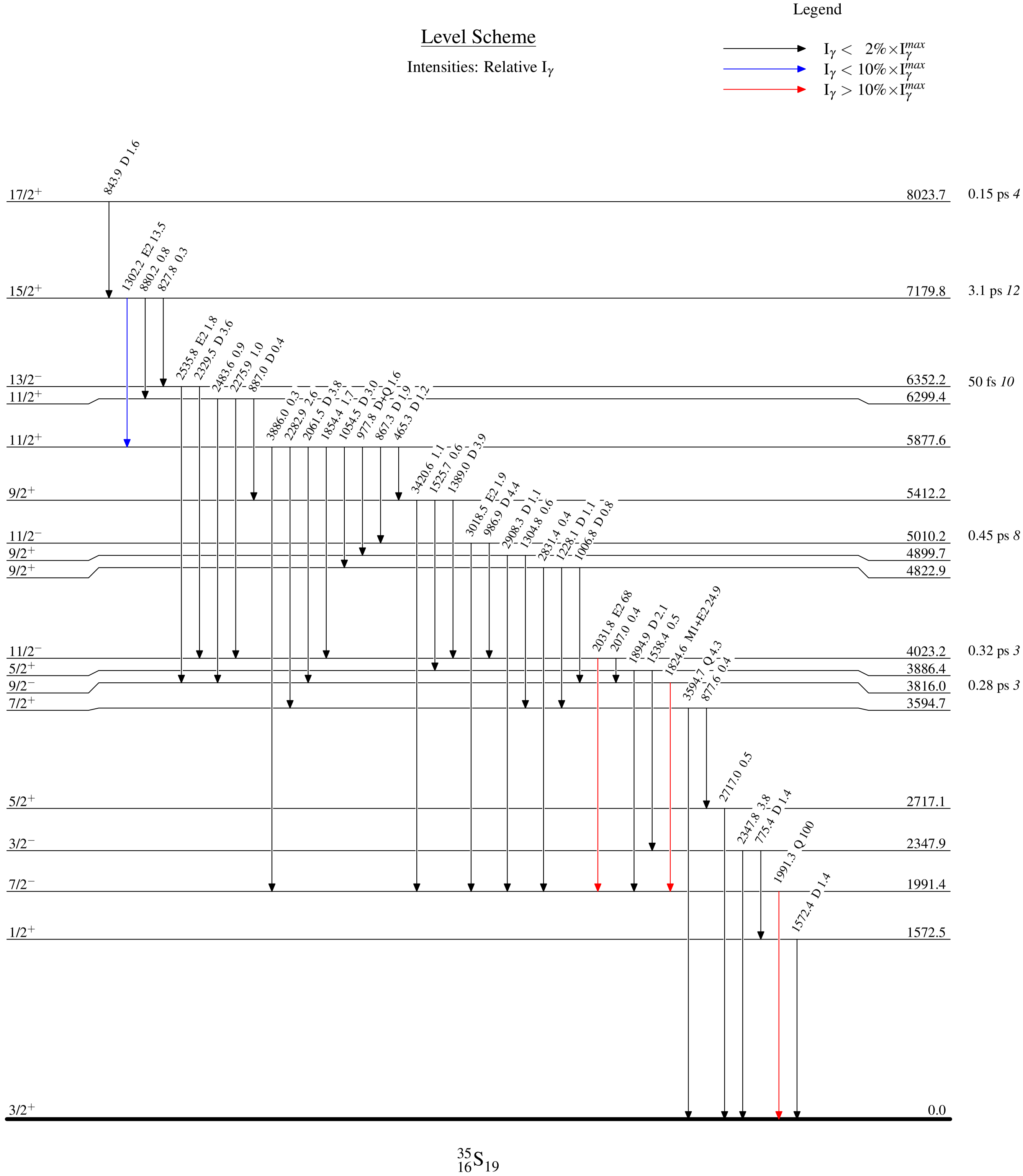
E_γ	I_γ	E_i(level)	J^π_i	E_f	J^π_f	Mult.†	δ†	Comments
1525.7 *8*	0.6 *1*	5412.2	9/2$^+$	3886.4	5/2$^+$			
1538.4 *8*	0.5 *2*	3886.4	5/2$^+$	2347.9	3/2$^-$			
1572.4 *3*	1.4 *4*	1572.5	1/2$^+$	0.0	3/2$^+$	D		R_{ADO}=0.88 *10*.
1824.6 *2*	24.9 *15*	3816.0	9/2$^-$	1991.4	7/2$^-$	M1+E2	+0.55 *9*	B(M1)(W.u.)=0.0099 *+14−12*; B(E2)(W.u.)=3.4 *+10−9* R_{ADO}=1.58 *5*, A_2=+0.48 *4*, A_4=+0.08 *6*. R_{ADO} seems questionable considering ΔJ=1 and M1+E2 assigned by the authors of 2014Ay01.
1854.4 *4*	1.7 *1*	5877.6	11/2$^+$	4023.2	11/2$^-$			
1894.9 *4*	2.1 *3*	3886.4	5/2$^+$	1991.4	7/2$^-$	D		R_{ADO}=0.69 *10*.
1991.3 *2*	100 *6*	1991.4	7/2$^-$	0.0	3/2$^+$	Q		R_{ADO}=1.45 *15*.
2031.8 *3*	68 *4*	4023.2	11/2$^-$	1991.4	7/2$^-$	E2		B(E2)(W.u.)=7.5 *+8−7* R_{ADO}=1.41 *10*, A_2=+0.29 *4*, A_4=−0.10 *6*.
2061.5 *3*	3.8 *2*	5877.6	11/2$^+$	3816.0	9/2$^-$	D		R_{ADO}=0.65 *9*.
2275.9 *6*	1.0 *2*	6299.4	11/2$^+$	4023.2	11/2$^-$			
2282.9 *4*	2.6 *1*	5877.6	11/2$^+$	3594.7	7/2$^+$			
2329.5 *9*	3.6 *7*	6352.2	13/2$^-$	4023.2	11/2$^-$	D		B(E1)(W.u.)=6.7×10^{-4} *+17−13*; B(M1)(W.u.)=0.023 *+6−5* R_{ADO}=0.8 *3*.
2347.8 *3*	3.8 *3*	2347.9	3/2$^-$	0.0	3/2$^+$			
2483.6 *8*	0.9 *2*	6299.4	11/2$^+$	3816.0	9/2$^-$			
2535.8 *11*	1.8 *2*	6352.2	13/2$^-$	3816.0	9/2$^-$	E2		B(E2)(W.u.)=5.3 *+17−11* R_{ADO}=1.5 *5*.
2717.0 *4*	0.5 *3*	2717.1	5/2$^+$	0.0	3/2$^+$			
2831.4 *9*	0.4 *1*	4822.9	9/2$^+$	1991.4	7/2$^-$			
2908.3 *5*	1.1 *2*	4899.7	9/2$^+$	1991.4	7/2$^-$	D		R_{ADO}=0.74 *14*.
3018.5 *7*	1.9 *2*	5010.2	11/2$^-$	1991.4	7/2$^-$	E2		B(E2)(W.u.)=0.223 *+49−39* R_{ADO}=1.35 *34*.
3420.6 *9*	1.1 *1*	5412.2	9/2$^+$	1991.4	7/2$^-$			
3594.7 *7*	4.3 *4*	3594.7	7/2$^+$	0.0	3/2$^+$	Q		R_{ADO}=1.36 *16*.
3886.0 *13*	0.3 *1*	5877.6	11/2$^+$	1991.4	7/2$^-$			

† From $\gamma(\theta)$, $\gamma\gamma(\theta)$(ADO) data, and RUL when $T_{1/2}$ is available in 2014Ay01.

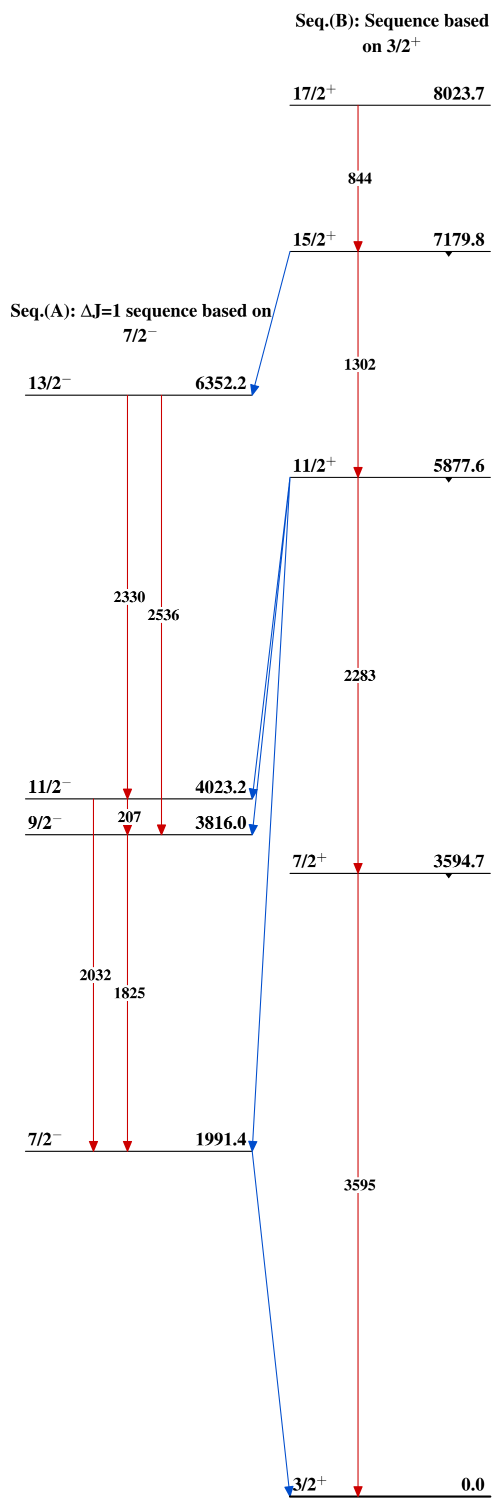
$^{24}Mg(^{14}N,3p\gamma)$ 2014Ay01

Level Scheme

Intensities: Relative I_γ



^{24}Mg(^{14}N,3pγ) 2014Ay01



$^{35}_{16}S_{19}$